



TENDINOPATHY: REHABILITATION

Comparing two eccentric exercise programmes for the management of Achilles tendinopathy. A pilot trial



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Summary *Objective:* To compare eccentric and static exercises as proposed by Stanish with eccentric exercises as proposed by Alfredson in the management of Achilles tendinopathy.

Methods: Patients with midportion Achilles tendinopathy for at least 3 months were included in this trial. They were sequentially allocated to receive either Stanish's exercise programme or Alfredson's exercise programme. Outcome measures were pain and function using the VISA-A score. Patients were evaluated at baseline, at the end of treatment (week 12), and 6 months (week 36) after the end of treatment.

Results: 41 patients met the inclusion criteria. At the end of treatment, there was a rise in VISA-A score in both groups compared with baseline ($p < 0.05$, paired t -test). There were significant differences in the VISA-A score between the groups at the end of treatment and at the 6-month follow up; Alfredson exercise programme group produced the largest effect ($p < 0.0005$, independent t -test).

Conclusion: An exercise programme based on Alfredson protocol was superior to Stanish model to reduce pain and improve function in patients with Achilles tendinopathy at the end of the treatment and at the follow-up. Further research is needed to confirm our results.

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Introduction

Achilles tendinopathy is a common clinical condition among athletes as well as in people who are non-athletes. This condition is characterized by the increased presence of

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fibroblasts, vascular hyperplasia, increased amounts of proteoglycans and glycosaminoglycans, disorganized collagen, absence of inflammatory cells and prostaglandin (Khan et al., 2000). The condition is degenerative rather than inflammatory and the term Achilles tendonitis is incorrect for clinical diagnosis (Rees et al., 2006). The term Achilles tendinosis refers to pathology of the Achilles tendon and is the best diagnostic term (Rees et al., 2006). On the other hand, the term Achilles tendinopathy refers to painful overuse tendon without implying pathology; it is ideal for clinical diagnosis (Kountouris and Cook, 2007).

The principal factors that lead to its development are extrinsic factors such as training errors, inappropriate footwear and intrinsic factors such as inflexibility, weakness and malalignment (Roos et al., 2004). Achilles tendinopathy typically presents with pain 2–6 cm proximal to the tendon insertion and this condition is called mid-portion Achilles tendinopathy (Kader et al., 2002). Achilles tendinopathy can also present in the Achilles tendon insertion, but insertional Achilles tendinopathy is less common than the midportion Achilles tendinopathy (Kountouris and Cook, 2007). An early symptom of Achilles tendinopathy is pain after exercise. As the pathological process progresses, pain occurs at the beginning of exercise and disappears during the exercise, later pain may occur during exercise, and, in severe cases, the pain interferes with activities of daily living (DiGiovanni and Gould, 1997).

However, no ideal treatment has emerged for the management of Achilles tendinopathy. Many clinicians advocate a conservative approach as the choice of treatment (Molnar and Fox, 1993) and physiotherapy is a conservative treatment that is usually recommended (Rees et al., 2006). A wide array of physiotherapy treatment options has been recommended for the management of Achilles tendinopathy such as electrotherapeutic modalities, exercise programmes, soft tissue manipulation, and manual techniques (Kountouris and Cook, 2007). While these treatments all have different theoretical mechanisms of action, they share the goals of reducing pain and improving function. Such a variety of treatment options suggest that the optimal treatment strategy is not known, and more studies are needed to determine the most effective treatment in patients with Achilles tendinopathy.

One of the most common physiotherapy treatments for Achilles tendinopathy is an exercise programme. One consisting of eccentric exercises has shown good clinical results in patellar and Achilles tendinopathy (Purdam et al., 2004; Jonsson and Alfredson, 2005; Young et al., 2005; Bahr et al., 2006) (Niesen-Vertommen et al., 1992; Alfredson et al., 1998; Mafi et al., 2001; Silbernager et al., 2001; Ohberg et al., 2004; Roos et al., 2004). On the other hand eccentric and static stretching exercises appear to be an effective treatment for tendinopathies (Stasinopoulos and Stasinopoulos, 2004; Manias and Stasinopoulos, 2006; Stasinopoulos and Stasinopoulos, 2006; Stasinopoulos et al., 2012). Such an exercise programme is used as the first treatment option for patients with Lateral Elbow Tendinopathy (LET) (Stasinopoulos and Johnson, 2004).

An exercise programme consisting of eccentric exercises in the management of Achilles tendinopathy was first proposed by Alfredson et al. (1998). On the other

hand, an exercise programme consisting of eccentric and static stretching exercises in the treatment of Achilles tendinopathy was first proposed by Stanish et al. (1986). To date, no study has made a direct comparison of the effect of exercise programme based on both protocol models. Therefore, the aim of the present article was to make a comparison of the effects of an exercise programme consisting of eccentric training (Alfredson and his coworkers model) and an exercise programme consisting of eccentric training and static stretching exercises (Stanish and his coworkers model) for the treatment of Achilles tendinopathy.

Methods

A controlled, monocentre trial was conducted in a clinical setting to assess the effectiveness of an eccentric training and an eccentric training with static stretching exercises. A parallel group design was used because cross-over designs are limited in situations where patients are cured by the intervention and do not have the opportunity to receive the other treatments after crossover (Johannsen et al., 1993). The subjects in one group have the opportunity to participate in the other protocol since it proves more beneficial.

Two investigators were involved in the study: (1) the primary investigator (DS) who administered the treatments and evaluated the patients to confirm the Achilles tendinopathy diagnosis, and (2) a physiotherapist (PM) who performed all baseline and follow-up assessments, and gained informed consent. All assessments were conducted by PM who was blind to the patients' therapy group. PM interviewed each patient to ascertain baseline demographic and clinical characteristics, including patient name, sex, age, duration of symptoms, previous treatment, occupation, affected leg and dominant leg. The study was approved by the manager of the clinic, who was the president of the ethical committee of the clinic.

Subjects between 35 and 55 years old who were experiencing pain in the Achilles region were examined and evaluated in a private outpatient physiotherapy clinic located in Athens between March 2002 and September 2010. The patients were either self-referred or referred by their physician or physiotherapist. All patients lived in Athens, Greece, were able to speak and understand English.

To be included in the study subjects had to have: (Kountouris and Cook, 2007; Rasmussen et al., 2008; Rompe et al., 2009).

1. tenderness with palpation 2–6 cm above the Achilles tendon insertion on the calcaneus (midportion Achilles tendinopathy)
2. minimum duration of symptoms three month
3. no history of trauma to the Achilles tendon
4. Unsuccessful conservative treatment before entering the study, but not in the preceding one month
5. No other conditions that could significantly contribute to posterior ankle pain (osteoarthritis, inflammatory arthritides, radiculopathy, systemic neurological conditions, etc)
6. No congenital or acquired deformities of the knee and ankle

Furthermore two of the below four tests had to be positive:

1. Positive heel-raise test (also known as calf-raise or toe-raise test) (the patient carries out plantar flexion from the standing position) (Rasmussen et al., 2008; Rompe et al., 2009)
2. Negative Thompson's (Simmonds') test (Examiner squeezes the calf musculature while observing for ankle plantar flexion) (Kountouris and Cook, 2007)
3. Positive painful arc sign (thickened portion of tendon moves with active plantar flexion and dorsiflexion of the ankle) (DiGiovanni and Gould, 1997)
4. Positive compression test (examiner compress Achilles tendon and the pain diminishes during plantar flexion and dorsiflexion) (Rees et al., 2006)

All patients received a written explanation of the trial before entry into the study and then gave signed consent to participate. They were allocated into one of two groups by sequential allocation. For example, the first patient with Achilles tendinopathy was assigned to the eccentric training and static stretching exercises group (Stanish and coworkers protocol), the second patient with Achilles tendinopathy to the eccentric training group (Alfredson and coworkers protocol), and so on.

All patients were instructed to use their ankles during the course of the study but to avoid activities that strained the ankle such as jumping, hopping and running. They were also to refrain from taking anti-inflammatory drugs or any other physiotherapy treatment throughout the course of study. Patient compliance with this request was monitored using a treatment diary.

Communication and interaction (verbal and non-verbal) between the therapist and patient was kept to a minimum, and behaviours sometimes used by therapists to facilitate

positive treatment outcomes were purposefully avoided. Patients for instance were given no indication of the potentially beneficial effects of the treatments or any feedback on their performance in the pre-application and post-application measurements (Vicenzino et al., 1996).

Stanish exercise programme

In the Stanish exercise programme, the patients performed a programme consisting of five steps. The first step was a general, whole-body warm up exercise not involving ankle plantar flexion. The second step was stretching exercises for the calf muscles. The stretching was a static stretch of gastrocnemius (knee in extension) and soleus (knee in flexion). The patients were instructed to hold these at least for 30 s and repeat each exercise three times. There was a 1-min rest between each stretch (Stasinopoulos and Stasinopoulos, 2004, 2006; Stasinopoulos et al., 2012). Next, 3 sets of 10 repetitions of the eccentric exercises were carried out once daily for six weeks and after six weeks, the patients were instructed to carry out 3 sets of 10 repetitions, three times per week for six more weeks. The intensity of the exercise should be such that pain, or discomfort, was experienced in the last set of 10 repetitions. This was done by having the patient stand on the edge of a step. The body weight was supported on the ball of the foot, so the heel was free. From that body position, the calf muscle was loaded eccentrically by having the patient lower the heel beneath the forefoot. The progression of eccentric training is shown in Table 1. Between each set there was a 2-min rest (Stasinopoulos and Stasinopoulos, 2004, 2006; Stasinopoulos et al., 2012). Every session ended with the same static stretch exercise as in the step 2. The patients were also instructed to use ice on the Achilles tendon for 5–10 min after the programme.

Table 1 Eccentric exercises based on Stanish protocol.

Week	Days	Exercise
1	1,2 3–5 6,7	Slow drop, bilateral weight support Moderate speed, bilateral weight support Fast drop, bilateral weight support
2	1,2 3–5 6,7	Slow, increased weight on symptomatic leg Moderate, increased weight on symptomatic leg Fast, increased weight on symptomatic leg
3	1,2 3–5 6,7	Slow, weight supported on symptomatic leg Moderate, weight supported on symptomatic leg Fast, weight supported on symptomatic leg
4	1,2 3–5 6,7	Slow, add 10% of body weight Moderate, add 10% of body weight Fast, add 10% of body weight
5	1,2 3–5 6,7	Slow, increase by 2.25–4.5 kg Moderate, increase by 2.25–4.5 kg Fast, increase by 2.25–4.5 kg
6	1,2 3–5 6,7	Slow, increase by 2.25–4.5 kg Moderate, increase by 2.25–4.5 kg Fast, increase by 2.25–4.5 kg
7–12	The same protocol every other day	The first set in slow speed, increase by 2.25–4.5 kg or more The second set in moderate speed, increase by 2.25–4.5 kg or more The third set in fast speed, increase by 2.25–4.5 kg or more

Alfredson exercise programme

In the Alfredson eccentric Achilles tendon programme the patients were instructed to do their eccentric exercises at a slow speed twice daily, 7 days/week, for 12 weeks. Two types of eccentric exercises were used. The calf muscle was eccentrically loaded both with the knee straight and, to maximise the activation of the soleus muscle, also with the knee bent. Each of the two exercises included 15 repetitions done in three sets. Between each set there was a 2-min rest (Stasinopoulos and Stasinopoulos, 2004, 2006; Stasinopoulos et al., 2012). The patients were told that muscle soreness during the first 1–2 weeks of training was to be expected. In the beginning, the loading consisted of the body weight, with the patients standing with 100% body weight on the affected leg. From an upright body position, the calf muscle was loaded by having the patient lower the heel beneath the forefoot. The subjects were only loading the calf muscle eccentrically; no concentric loading was used. Instead, the noninjured leg was used to return to the start position. The patients were told to go ahead with the exercise even if they experienced pain. However, each subject was told to stop the exercise if the pain became disabling. If the subjects could perform the eccentric loading exercise without experiencing any minor pain or discomfort, they were instructed to increase the load by adding weight. This was done by using a backpack that was successively loaded with weight.

Outcome measures

Pain, function and drop out rate were measured in the present study. The above-mentioned 4 special tests were not used as outcome measures because the validity and reliability of these tests as outcome measures is unknown. Each patient was evaluated at the baseline (week 0), at the end of treatment (week 12) and at 6 months (week 36) after the end of treatment.

The VISA-A questionnaire was used to monitor the pain and function of patients. The instrument is a simple questionnaire, consisting of eight questions. This simple questionnaire takes less than 5 min to complete and once patients are familiar with it they can complete most of it themselves. Patients rate the amount of pain performing some tasks such as climbing stairs, squatting. The perfect score is 100. It is a valid and reliable outcome measure for patients with Achilles tendinopathy (Robinson et al., 2001).

A drop-out rate was also used as an indicator of treatment outcome. Reasons for patient drop-out were categorized as follows: (1) withdrawal without reason, (2) not returned for follow-up, and (3) request for an alternative treatment.

Analysis

The change from baseline was calculated for each follow-up. Differences between groups were determined using the independent *t*-test. The difference within groups between baseline and end of treatment was analysed with a paired *t*-test. A 5% ($p = 0.05$) level of probability was adopted as the level of statistical significance. SPSS 11.5 statistical software was used for the statistical analysis (SPSS Inc., Chicago, IL, USA).

Results

Sixty seven patients eligible for inclusion visited the clinic within the trial period. Fourteen were unwilling to participate in the study, and twelve did not meet the inclusion criteria described above. The other 41 patients were sequentially allocated to one of the two possible groups: (a) Stanish exercise programme ($n = 21$; mean (SD) age 48.44 (5.12) years); (b) Alfredson exercise programme ($n = 20$; mean (SD) age 48.24 (5.09) years). Patient flow through the trial is summarised in a CONSORT flow chart (Fig. 1).

The mean age of the patients was 48 years, and the duration of Achilles tendinopathy was about 7 months (Table 2). Achilles tendinopathy was in the dominant leg in 90% of patients (Table 2). There were no significant differences in mean age ($p > 0.05$, independent *t* test) or the mean duration of symptoms ($p > 0.05$, independent *t* test) between the groups. Patients had received a wide range of previous treatments (Table 3). Drug therapy had been tried by 75%. All patients were recreational athletes.

VISA-A

Baseline VISA-A score was 37 (95% CI 26–49) for the whole sample ($n = 41$; Table 4). There were no significant differences between the groups for baseline VISA-A score ($p > 0.05$ independent *t* test; Table 3). At week 12, there was a rise in VISA-A score of 40 units in the Alfredson exercise programme group and 25 units in the Stanish exercise programme group with the baseline ($p < 0.05$, paired *t* test; Table 5). There were significant differences in the magnitude of improvement between the groups at weeks 12 and 36 ($p < 0.05$ independent *t* test; Table 4).

Drop outs

There were no drop outs, no adverse effects were reported and all patients successfully completed the study.

Discussion

Comparison of outcomes

The results obtained from this controlled clinical trial are novel; as to date, there have been no data comparing the effectiveness of Stanish exercise programme and Alfredson exercise programme for the reduction of pain and improvement of function in Achilles tendinopathy in recreational athletes between 35 and 55 years old. The protocol that was proposed by Alfredson produced a better effect at the end of the treatment and 6 months after the treatment ended.

In the Alfredson exercise programme the eccentric calf muscle training reduced pain and improved function more than the Stanish exercise programme because in the first protocol the patients exercised both calf muscles (gastrocnemius and soleus) only eccentrically, with more sets and with more repetitions everyday for the same treatment period. Moreover, the load of eccentric exercises in the Alfredson exercise protocol was increased according

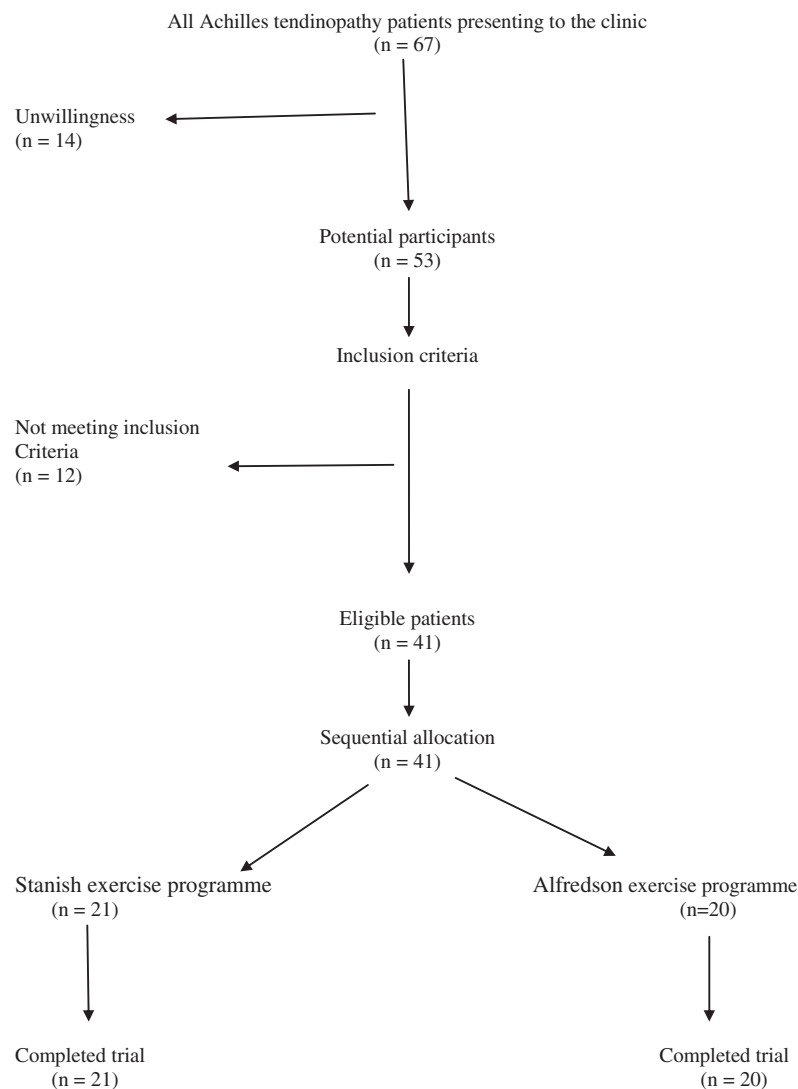


Figure 1 Flow chart of the study.

to the patients' symptoms otherwise the results are poor (Jensen and Di Fabio, 1989) and in the Stanish exercise programme this factor was not followed. Furthermore, eccentric exercises in the Alfredson exercise protocol were performed at a low speed in every treatment session because this allows tissue healing (Kraushaar and Nirschl, 1999) and this issue was also ignored in the Stanish exercise protocol. In the Stanish exercise programme ice was recommended at the end of the treatment but research has shown that ice as a supplement to an eccentric exercise programme offers no benefit to patients with tendinopathy (Manias and Stasinopoulos, 2006). Finally, the avoidance of painful activities is crucial for tendon healing, because training during the treatment period increases patients' symptoms and delays tendon healing (Visnes et al., 2005) and this issue was followed in both exercise programs.

Standard eccentric exercises as described by Alfredson offer adequate rehabilitation for tendon disorders, but some patients with tendinopathies do not respond to this prescription alone (Cannell et al., 2001). For this reason, clinicians combine eccentric exercises with static

stretching exercises as described by Stanish in the management of tendinopathies. Studies have shown positive results in the treatment of tendon injuries using eccentric training as described by Alfredson and static stretching exercises as described by Stanish (Manias and Stasinopoulos, 2006; Stasinopoulos and Stasinopoulos, 2004, 2006; Stasinopoulos et al., 2012). The results of the present study might be better if the patients who underwent the Alfredson eccentric exercise protocol also

Table 2 Demographics data.

	Age (years (SD))	Duration of symptoms (months)	Baseline VISA-A
Stanish exercise programme	48.44 (5.12)	6.9	38 (95%CI 28–48)
Alfredson exercise programme	48.24 (5.09)	7.1	36 (95%CI 27–47)
p value	>0.05	>0.05	>0.05

Table 3 Previous treatments of participants.

	Stanish exercise programme (%)	Alfredson exercise programme (%)
Drugs	15 (71.4)	14 (70)
Physiotherapy	2 (9.6)	3 (15)
Injection	4 (19)	3 (15)

Values are number (%).

undertook static stretching exercises for gastrocnemius and soleus as described by Stanish before and after the eccentric exercise programme.

Eccentric training and static stretching exercises appear to reduce pain and improve function, reversing the pathology of tendinopathy (Hawary et al., 1997; Khan et al., 2000, 2002; Ohberg et al., 2001) as supported by experimental studies on animals (Vilarta and Vidal, 1989). The way that eccentric training and static stretching exercises achieves the goals remains uncertain as there is a lack of good quality evidence to confirm that physiological effects translate into clinically meaningful outcomes and vice versa.

There are two types of exercise programs: home exercise programs and exercise programs carried out in a clinical setting. A home exercise program is commonly advocated for patients with tendinopathies such as Achilles tendinopathy because it can be performed any time during the day without requiring supervision by a physiotherapist. Such a home exercise programme was administered in the present controlled clinical trial. Our clinical experience, however, has shown that patients fail to comply with the regimen of home exercise programs (Stasinopoulos and Johnson, 2004). This problem can be solved by exercise programs performed in a clinical setting under the supervision of a physiotherapist. For the purpose of this report, "supervised exercise program" refers to such programs.

Previous trials have found that a home exercise program reduced the pain in patellar tendinopathy (Purdam et al., 2004; Jonsson and Alfredson, 2005; Young et al., 2005; Bahr et al., 2006), lateral elbow (Pienimäki et al., 1996), and Achilles tendinopathy (Niesen-Vertommen et al., 1992; Alfredson et al., 1998; Silbernager et al., 2001; Mafi et al., 2001; Ohberg et al., 2004; Roos et al., 2004). However, it was performed for about 3 months in all previous studies. In contrast, in the studies of Stasinopoulos and colleagues a supervised exercise program was administered for 1 month (Stasinopoulos and Stasinopoulos, 2004; Manias and Stasinopoulos, 2006; Stasinopoulos and Stasinopoulos, 2006; Stasinopoulos et al., 2012). Thus it seems that the supervised exercise program may give good long-term clinical results in

Table 5 Change in VISA-A from baseline.

	Week 12	Week 36
Alfredson exercise programme	40	42
Stanish exercise programme	25	26
<i>p</i> value	<0.05	<0.05

Values are mean scores *p* Values for independent *t* test on change in VISA score from baseline are shown.

a shorter period of time than a home exercise program. The most likely explanation for this difference is that a supervised exercise program achieves a higher degree of patient compliance. Studies to compare the effects of these two types of exercise programs are required to confirm the findings of the present controlled clinical trial.

However, this trial does have some shortcomings. First, a power analysis was not performed. Second, although this study was not a randomised controlled trial because a genuine randomisation procedure was not followed, the use of sequential allocation of patients to treatment groups allowed a true cause and effect relation to be demonstrated. Third, no placebo (sham) or no treatment group was included in the present trial. The placebo (sham)/no treatment group is important when the absolute effectiveness of a treatment is determined. However, the absolute effectiveness of technique based interventions is difficult to investigate because a good and trustworthy placebo (sham)/no treatment control for exercise programmes appears to be difficult or impossible to devise, due in part to difficulties in defining the active element of these treatments. Absolute effectiveness also does not provide the therapists with information as to which is the most appropriate treatment for the management of a condition, in this case Achilles tendinopathy. Finally, the blinding of patients and therapists would be problematic in that case, if not impossible, because patients know if they are receiving the exercise programme treatment and therapists need to be aware of the treatment to administer it appropriately.

In addition to the weaknesses discussed, structural changes in the tendon related to treatment interventions were not shown due to the lack of diagnostic ultrasound; plyometric training of the patients were not used in the present trial, because this was out of the aim of the study. The role of plyometric training in Achilles tendinopathy patients will be assessed in a future trial. Improvement of function and reduction of pain occurred in recreational athletes who were between 35 and 55 years old, but it is unknown if the exercise programme is an effective treatment approach in all patients with Achilles tendinopathy according to sex, age and occupation. Further research is needed to establish the effectiveness of an exercise programme in the management of Achilles tendinopathy.

Conclusion

The Alfredson exercise programme consisting of eccentric exercises reduced pain and improved function in recreational athletes between 35 and 55 years at the end of the treatment and at follow-up. A combination of eccentric exercises with static stretching exercises, might produce better results. Supervised exercise programmes might be

Table 4 VISA-A: self-score completed prior to examination.

	Week 0	Week 12	Week 36
Stanish exercise programme	38 (95%CI 28–48)	63 (95%CI 57–73)	64 (95%CI 58–79)
Alfredson exercise programme	36 (95%CI 27–47)	76 (95%CI 77–90)	78 (95%CI 75–94)

superior to home exercise programmes. Controlled studies are needed to establish the effects and the mechanism of action of such exercise programmes in patients with Achilles tendinopathy.

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